Mindmatrix Project no 38

Parisara-Cycle

Green Commuter Guide — Android Application

Full Project Documentation

Name: Ayesha Bee
Usn: 1CD22IS021

College: Cambridge Institute of Technology

Email id:ayeshabee.22ise@cambridge.edu.in

Project Name: Parisara-Cycle
Platform: Android (GenAI-Powered)
Target Audience: Students and Daily Workers in Tier-2 and Tier-3 Towns
Document Type: Full Project Specification
Version: 1.0

1. Project Overview

Parisara-Cycle is an Android application designed as a community-driven Green Commuter Guide for students and workers in growing Indian towns. The core idea is to make cycling a safer, smarter and more rewarding way to commute — not just a last resort. The name 'Parisara' means 'environment' in Kannada, reflecting the app's roots in sustainability and local culture.

The app combines cycle-friendly route navigation, real-time community hazard mapping, pit-stop discovery and a buddy-riding system to address the practical barriers that keep people off their cycles every day.

1.1 Background

In many tier-2 and tier-3 towns across India, public transport is unreliable and roads are designed almost entirely for motorized vehicles. Cyclists are forced to share space with fast-moving traffic on national highways, often with no designated lane or safe alternative. This is a common reality in towns like Mandya, Tumkur, Hassan and similar regions where distances between home, school and workplace are short enough to cycle but roads feel too dangerous to try.

Most people in these towns own a cycle or can access one cheaply. The missing piece is not the cycle — it is confidence in the route and safety in numbers. Parisara-Cycle aims to fill that gap.

1.2 What the App Does

- Maps safe, low-traffic cycling routes using the Google Maps Bicycling layer
- Lets users report and view road hazards like potholes, broken paths and dangerous junctions
- Shows nearby cycle repair shops and water points so riders are never stuck
- Tracks CO2 savings for every ride and shows a running monthly total
- Connects nearby cyclists so they can ride together for safety

2. Problem Statement

Despite cycling being a practical and affordable mode of transport in smaller Indian towns, adoption remains low. Through observation and user research, three core problems stand out:

Problem 1 — Road Safety Anxiety

Cyclists in growing towns share roads with buses, trucks and fast bikes, often without any cycle lane or safe shoulder. Students especially feel that cycling to college is unsafe because of reckless driving on the stretches connecting residential areas to main roads. This fear is not irrational — it is backed by real accident data on state highways.

Problem 2 — No Knowledge of Safer Inner Routes

Most towns have a network of internal roads — village lanes, service roads, and colony paths — that are much quieter than the main road. But these routes are not on Google Maps, not well-known, and often not shared between communities. A newcomer to a town has no way to discover that there is a perfectly good shaded side-lane that cuts through the market, avoids the highway and gets them to college in the same time.

Problem 3 — No Way to Report Cycle Path Problems

A deep pothole at a specific turn, a broken drain cover on a lane, a construction blockage on the inner road — these affect cyclists the most, but there is no existing mechanism to report them, track them or alert other riders. By the time word spreads informally, dozens of people may have already had a fall.

3. Scope of the Application

3.1 In-Scope Features

The following are part of the app's planned functionality:

- Cycle-specific route navigation avoiding national and state highways
- Turn-by-turn navigation with audio guidance
- Community hazard pinning — potholes, dangerous intersections, blocked paths
- Hazard display on map with category icons and tap-to-view details
- Upvoting of existing hazard reports to build credibility
- Pit-stop finder for cycle repair shops and drinking water points via Google Places API

- CO2 savings tracker — per ride, daily, weekly and monthly totals
- Monthly totals persisted in Firebase so they survive reinstallation
- Buddy System — live location sharing, finding nearby cyclists, join-ride requests
- User profile with cumulative ride stats
- Login via Google account or email/password
- Lightweight UI optimized for mid-range Android devices

3.2 Out-of-Scope (Phase 1)

To keep the first version focused and deliverable, the following are excluded from Phase 1:

- iOS version or cross-platform build
- Navigation for cars or motorcycles
- Fully offline maps or downloadable route packs
- Payment, rewards or e-commerce features
- Integration with official government or smart-city data feeds
- Advanced AI route personalization (planned for Phase 2)

4. Functional Requirements

This section describes what the application must do. Each requirement is written from the user's perspective.

4.1 Authentication

FR-01: Login and Registration

- The app must support login via Google Sign-In (OAuth 2.0)
- Users must also have the option to register with an email and password
- On first login, users complete a short onboarding: set their name, choose their home area and select route preference (fastest vs safest)
- Login sessions must persist across app restarts using secure token storage

4.2 Route Navigation

FR-02: Safe Route Calculation

- The app calculates cycling routes using Google Maps Directions API in 'bicycling' mode
- Routes that pass through national or state highways must be filtered out where alternate paths exist
- Users choose between three route styles: Fastest, Safest, or Shaded (prefers tree-lined lanes)
- Navigation includes turn-by-turn audio directions
- Hazard pins from the community are shown on the map while navigating

4.3 Hazard Reporting

FR-03: Pin a Danger Zone

This is one of the most important features of the app. Any logged-in user should be able to mark a problem on the map so other cyclists know about it.

- Long-pressing any point on the map opens a hazard report form
- The report includes: hazard type (dropdown — pothole / blocked path / dangerous junction / other), a short text description and an optional photo
- GPS coordinates are filled automatically from the device's location
- Submitted reports appear instantly on the community map via Firebase Firestore
- Other users can upvote a hazard report to mark it as verified
- Hazard pins with no upvotes after 30 days are automatically archived

4.4 Pit-Stop Finder

FR-04: Nearby Cycle Services

- Users can tap the 'Pit-Stop' button to search for nearby cycle repair shops, puncture stalls and water points
- Results are displayed both on the map and in a scrollable list sorted by distance
- Data is pulled from Google Places API filtered by business category
- Users can suggest a new pit-stop location (submitted for community moderation)

4.5 Eco-Stats

FR-05: CO2 Savings Tracking

The logic is straightforward: every kilometre cycled instead of riding a petrol vehicle saves approximately 120 grams of CO_2. This number is based on the average emission factor of petrol two-wheelers commonly used in India.

- After each ride, the app calculates: Distance (km) x 120 = CO_2 saved in grams
- The dashboard shows today's savings, this week's total, and the current month's total
- Monthly totals are stored in Firebase Firestore and do not reset on app reinstall — they are tied to the user's account
- A simple bar or stat card shows the user's running monthly total

4.6 Buddy System

FR-06: Find a Cycling Partner

- Users can opt in to share their live location with nearby app users
- Other cyclists within a set radius appear as markers on the map
- A user can send a 'Join My Ride' request to a nearby cyclist
- If accepted, both users see each other's real-time location and estimated arrival time
- Location data is stored only in Firebase Realtime Database and deleted when the ride session ends
- Location sharing stops automatically after 15 minutes of no movement

5. Non-Functional Requirements

5.1 Performance

Requirement	Target
App cold start (home screen load)	Under 3 seconds on a mid-range Android device
Map initial load on 4G	Under 3 seconds
Route calculation response	Under 5 seconds from tap to result
Hazard report upload to Firestore	Under 2 seconds
Background GPS usage (Buddy System)	Less than 5% battery per hour

5.2 Security and Privacy

- All data transfers must use HTTPS / TLS 1.3
- Live location shared in the Buddy System must not be logged to any server — only stored in memory during the active session
- Firebase Security Rules must enforce that users can only read and write their own profile data
- Hazard reports are anonymous by default — no username shown on the pin
- The app must comply with India's Digital Personal Data Protection Act, 2023 (DPDPA)

5.3 Reliability

- Core features (map view and navigation) should work 99.5% of the time
- The last-cached route must be viewable even without an internet connection
- Firebase should be configured with multi-region replication to prevent data loss

5.4 Usability

- The interface should follow Material Design 3 guidelines and feel familiar to Android users
- Minimum touch target size of 48dp for all interactive elements
- Color contrast must meet WCAG 2.1 Level AA for accessibility
- App should support English as default, with Kannada as the first regional language

5.5 Code and Maintainability

- MVVM (Model-View-ViewModel) architecture with a Repository pattern
- Minimum 70% unit test coverage for core business logic
- API keys must never be committed to version control — stored in environment config files
- All modules must be independently testable

6. Features

6.1 Must-Have Features (Phase 1 — MVP)

These features are non-negotiable for the first release. Without them, the app does not solve the core problem.

Feature	What It Does	Why It's Must-Have
Safe Route Map	Cycle-specific navigation using Google Maps bicycling mode, avoids highways	Core purpose of the app — without this, the safety problem is unsolved
Hazard Pin Reporting	Long-press map to report potholes, blocked paths, dangerous junctions	Community data is the only way to surface unlisted road problems
Hazard Overlay on Map	All reported pins shown with color-coded icons; tap to see details	Users need to see the data to benefit from community reporting
Eco-Stats Dashboard	CO_2 saved per ride shown as daily and monthly totals	Turns cycling into a proud green choice — core to the app's identity
Monthly Eco-Stats Persistence	Monthly totals backed up in Firebase, not lost on reinstall	Explicitly listed as a success criterion — must be delivered
Buddy System (Basic)	Live location sharing, view nearby cyclists, send join-ride requests	Safety-in-numbers addresses the key fear of riding alone
Pit-Stop Finder	Nearby repair shops and water points via Google Places	Practical blocker — knowing help is nearby builds confidence
User Login	Google Sign-In or email/password via Firebase Auth	Needed to persist data and associate reports with accounts
Battery-Efficient UI	No heavy animations, optimized background processes, minimal wake locks	Explicitly listed as a success criterion for mid-range device support

6.2 Good-to-Have Features (Phase 2 and Beyond)

These features add significant value but are not needed for the first launch. They are listed here so they can be designed for from the start even if not built yet.

Feature	Description	Phase
Ride Recording	Record a full ride with GPS track, duration, distance and post-ride summary	Phase 2
Leaderboard	Monthly CO_2 leaderboard for users in the same locality	Phase 2
Shaded Route Algorithm	Prefer tree-lined and covered roads based on satellite greenery	Phase 2

Feature	Description	Phase
	data	
AI Hazard Photo Check	Use Gemini API to verify that uploaded hazard photos match the reported category	Phase 2
Badges and Milestones	Unlock badges for ride counts, CO2 saved and hazards reported	Phase 2
Community Pit-Stop Submissions	Users suggest new repair shops; moderator or upvote-based approval	Phase 2
Buddy Group Chat	Basic text chat between two cyclists on an active shared ride	Phase 2
More Regional Languages	Hindi, Tamil, Telugu support in addition to Kannada	Phase 2
Offline Route Cache	Download a route to use without internet; view cached hazard map offline	Phase 2
Smart Notifications	Push alerts for new hazards near a saved route or incoming buddy requests	Phase 3
Google Fit Integration	Sync ride data to show calories burned alongside CO2 saved	Phase 3
Smart City API Integration	Pull official road condition data from government smart city platforms	Phase 3

7. Technical Implementation

7.1 Technology Stack

Area	Technology
Language	Kotlin
Architecture	MVVM with Repository Pattern
UI	Jetpack Compose / XML with Material Design 3
Maps and Navigation	Google Maps Android SDK + Directions API (bicycling mode)
Places / Pit-Stops	Google Places API
User Auth	Firebase Authentication (Google Sign-In, Email/Password)
Database (Hazards, Eco-Stats, Profiles)	Firebase Cloud Firestore
Real-Time Location (Buddy System)	Firebase Realtime Database

Area	Technology
Photo Storage	Firebase Storage
AI / GenAI Features	Google Gemini API (Phase 2 hazard verification)
Push Notifications	Firebase Cloud Messaging (FCM)
Build System	Gradle with Kotlin DSL
Version Control	Git and GitHub

7.2 CO2 Savings Calculation

The formula used to calculate environmental impact per ride is:

$$\text{CO2 Saved (g)} = \text{Distance Cycled (km)} \times 120$$

The figure of 120 grams per kilometre represents the average CO2 emission of a petrol two-wheeler (scooter or motorcycle) in India under typical urban driving conditions. For example, a 4 km ride to college and back saves approximately 960 grams — nearly 1 kg of CO2 — on that single day.

This value is stored against the user's account after every ride and aggregated into daily, weekly and monthly totals in Firestore.

7.3 User Flow Summary

1. User opens the app and logs in with Google or email
2. Home screen shows a map centered on their current location
3. User enters a destination and taps 'Find Route'
4. App shows route options: Fastest / Safest / Shaded with estimated CO2 savings for each
5. Navigation starts with turn-by-turn audio; community hazard pins are visible on the route
6. During the ride, user can optionally long-press to report a hazard or tap to find a nearby pit-stop
7. If Buddy System is enabled, nearby cyclists appear on the map; user can send a Join Ride request
8. On reaching the destination, the app shows the ride summary: distance, time and CO2 saved
9. CO2 data is saved to the user's Eco-Stats dashboard automatically

8. Impact Goals

Parisara-Cycle is not just a navigation app. The larger goal is to make cycling culturally normal and safe in Indian towns where it has been seen either as a poor person's last resort or a fitness hobby for the privileged. The app tries to shift that perception by making every cyclist feel safer, more informed and proud of the environmental contribution they're making.

Net Zero and Climate Action

Promoting carbon-neutral daily transport in tier-2 and tier-3 towns directly contributes to India's net-zero commitments. Even a 10% shift in short-distance commutes from petrol vehicles to cycles in a town of 100,000 people would amount to thousands of tonnes of CO_2 saved annually.

Public Health

Cycling 4-6 km a day provides meaningful cardiovascular exercise. Widespread adoption in towns would reduce sedentary lifestyle diseases — particularly important in areas with limited gym infrastructure and rising rates of diabetes and hypertension among young people.

Road Safety

The hazard reporting system creates a data layer that does not currently exist for cyclists. Over time, this data could inform local bodies about which roads need repair priority or infrastructure investment — starting from the ground up, built by the people who actually use those roads.

Community and Belonging

The Buddy System is designed to reduce the social isolation that often comes with cycling in towns. When students can see that other students from their area are also cycling to the same college and can find a riding partner with one tap, it normalises the behaviour and makes it feel safe and social.

9. Success Criteria

The following criteria define whether the app has been built correctly and meets its objectives. These are the benchmarks that will be checked during evaluation.

Eco-Stats Monthly Persistence

The monthly CO_2 total must be stored in Firebase and restored correctly even if the user uninstalls and reinstalls the app. This is a hard requirement. The stats are tied to the user's account, not the device.

Hazard Pinning Works End-to-End

Any logged-in user must be able to long-press the map, fill the hazard form, and see the pin appear on the map within 2 seconds. The pin must be visible to other users immediately. This should be tested on at least two different devices simultaneously.

UI is Lightweight and Battery-Efficient

The app must be tested on a mid-range device like a Redmi 10 or equivalent (3-4 GB RAM). During a 30-minute ride simulation with GPS active, background battery consumption must stay below 5% per hour.

Route Avoids Highways

When a route exists via an inner road and a highway, the app's 'Safest' mode must choose the inner road. This must be manually verified for at least three real-world route pairs in the target town.

Buddy System Response Time

When two users are within the configured radius and one sends a Join Ride request, the other must receive it within 5 seconds. Both users must see each other's location markers on the map within that time.

10. Risks and How We Plan to Handle Them

Risk	Mitigation
Google Maps API quota runs out with heavy testing	Cache recent routes in Firestore; limit re-requests for the same origin-destination pair within a session
Firebase Realtime DB is slow in rural areas with 2G/3G connectivity	Show last-known buddy location with a 'last updated' timestamp so users know the data may be stale
Fake or spam hazard reports flood the map	Require login to report; community upvote/downvote system; auto-archive pins with zero engagement after 30 days
Continuous GPS drains battery significantly	Use Android fused location provider with adaptive polling; auto-pause when user is stationary for 15 minutes
Low initial adoption — no community data makes the map feel empty	Seed initial hazard data manually for the launch area; partner with local colleges for early user base
API keys accidentally pushed to GitHub	Use local.properties and .env files; add to .gitignore; use GitHub secret scanning alerts

End of Document

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